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Hasse-Weil Theorem

p prime, $q = p^n$

$$\Phi : \bar{\mathbb{F}}_q \to \bar{\mathbb{F}}_q = \bar{\mathbb{F}}_p = \bigcup_n \mathbb{F}_{p^n}$$

$$\Phi(x) = x^q$$

it is a field homomorphism. Induces a map for $E/\mathbb{F}_q$

$$\Phi : E(\bar{\mathbb{F}}_q) \to E(\bar{\mathbb{F}}_q)$$

$$\Phi(x, y) = (x^q, y^q)$$

Frobenius is compatible wih group structure on $E(\bar{\mathbb{F}}_q)$.

Definition: Isogeny

E, E' are EC on K. An isogeny $\alpha : E \to E'$ is a rational map such that the induced map

$$E(\bar{K}) \to E'(\bar{K})$$

is a group homomorphism

Example: Frobenius

Isogeny $\alpha : E \to E$ is an endomorphism.

If $\alpha : E/K \to E'/K$ is an isogeny then

$$\alpha : E(L) \to E'(L)$$

for $K \subseteq L \subseteq \bar{K}$ is an isogeny.

$$E(L) \subseteq E(\bar{K})$$

Example

Let E/K be any EC, for all n multiplication by n is an endomorphism.

$$[n] : E \to E$$

$$P \to nP$$

Everything we do is polynomials and it preserves group structure.

Recall:

An isogeny $\alpha : E \to E'$ viewed as a rational map, has a canonical form.

$$\alpha(x, y) = (r_1(x), yr_2(x))$$

where $r_1(x) = \frac{p(x)}{q(x)}, r_2(x) = \frac{u(x)}{v(x)}$ and each quotient is reduced, so no common factors over $\bar{K}$.

If $q(x) = 0$ for some $x, y \in E(\bar{K})$, then we set $\alpha(x, y) = 0_{E'}$ and otherwise we showed $v(x) \neq 0$ and hence α is well defined.

Def

Let $\alpha : E/K \to E'/K$ be an isogeny.

1. The degree of α is $\deg(\alpha) = \max\{\deg(p), \deg(q)\}$.
2. α is called separable if the formal derivative $r_1'(x)$ is not identically zero $\quad p(x)q'(x) - p'(x)q(x) \neq 0$

$$\Phi_q = \alpha : E(\bar{\mathbb{F}}_q) \to E(\bar{\mathbb{F}}_q)$$

$$\infty \to \infty$$

$$(x, y) \to (x^q, y^q) \in E(\bar{\mathbb{F}}_q)$$

$$(y^q)^2 = (x^q)^3 + Ax^q + B$$

$$(y^2)^q = (x^3 + Ax + B)^q$$

Is Φ_q separable?

$$(x^q)' = qx^{q-1} = 0 \text{ in } \mathbb{F}_q$$

so it is not separable.

Prop

Let $\alpha : E \to E'$ be a nonzero isogeny. If α is separable then

$$\#\ker(\alpha : E(\bar{K}) \to E'(\bar{K})) = \deg(\alpha)$$

and otherwise $\#\ker(\alpha) < \deg(\alpha)$

Observe $\#E(\mathbb{F}_q) = \#\ker(\alpha)$

For $E/\mathbb{F}_q$

$$\alpha : \Phi_q^n - \mathrm{id} : E \to E$$
$$P \to \Phi_q^n(P) - P$$
$$\ker(\alpha : E(\bar{\mathbb{F}}_q) \to E(\bar{\mathbb{F}}_q)) = \#E(\mathbb{F}_{q^n})$$

(or without n easier)

For $E/\mathbb{F}_q$

$$\alpha : \Phi_q - \mathrm{id} : E \to E$$
$$P \to \Phi_q(P) - P$$
$$\ker(\alpha : E(\bar{\mathbb{F}}_q) \to E(\bar{\mathbb{F}}_q)) = \#E(\mathbb{F}_q)$$
$$P \in \ker(\alpha) \Leftrightarrow \Phi_q(P) - P = \infty$$
$$\Leftrightarrow \Phi_q(P) = P$$

we saw that these points P are exactly $E(\mathbb{F}_q)$

The only points frobenius acts as identity is those in $\mathbb{F}_q$, so only unchanged points are in the kernel. In higher extensions, frobenius doesn't act as the identity.

Proof

Since $\alpha \neq 0$ and is a group homomorphism on $E(\bar{K}) \to E'(\bar{K})$ it is non-constant.

Thus $\alpha : E(\bar{K}) \to E'(\bar{K})$ is surjective. Let $Q = (a, b) \in E'(\bar{K})$ and $P = (x_0, y_0) \in E(\bar{K})$.

Exercise: Show the prop on surjectivity generalizes to the case of $E \to E'$

Since $E'(\bar{K})$ is infinite we can choose Q st

1. $a, b \neq 0$
2. $\deg(p - qa) = \max\{\deg(p), \deg(q)\} = \deg(\alpha)$

the only case in which $\deg(p-qa) < \deg(\alpha)$ is when $\deg(p) = \deg(q)$ and their leading coefficients λ, δ respectively satisfy

$$\lambda - a\delta = 0 \Leftrightarrow a = \frac{\lambda}{\delta}$$

so we choose Q such that $a \neq \frac{\lambda}{\delta}$.

Since $\deg(p - aq) = \deg(\alpha)$, $p(x) - aq(x)$ has exactly $\deg(\alpha)$ roots over $\bar{K}$ (possibly repeated roots).

We claim that the number of distinct roots of $p - aq$ is exactly the number of sources P of Q (under α).

Since $(a, b) \neq (\infty, \infty)$, then

$$r_1(x_0) \neq 0 \Leftrightarrow q(x_0) \neq 0$$

since $b \neq 0$ and we have

$$y_0 r_2(x) = b$$

we have $y_0 = b/r_2(x_0)$, so y_0 is completely determined by x_0.

So it is enough to count the x_0's which in turn must satisfy $\frac{p(x_0)}{q(x_0)} = a$

$$\Leftrightarrow p(x_0) - aq(x_0) = 0$$

i.e the roots of $p - aq$

Since α is a group homomorphism on $E(\bar{K}) \to E'(\bar{K})$, then $\#\ker(\alpha)$ is the same as the number of sources of any given point $Q \in E'(\bar{K})$

Which is enough to analyze the number of distinct roots x_0 of $p - aq$.

x_0 is a repeated root of $p - aq \Leftrightarrow p(x_0) - aq(x_0) = 0$ and also $p'(x_0) - aq'(x_0) = 0$. Multiply both equations to get

$$ap(x_0)q'(x_0) = ap'(x_0)q(x_0)$$

Since $a \neq 0$

$$p(x_0)q'(x_0) - p'(x_0)q(x_0) = 0$$
$$r_1'(x_0) = 0$$

by the quotient rule applied to r_1'.

If α is not separable

$$r_1'(x) = 0$$

which means $p - aq$ has repeated roots and $\#\ker(\alpha) < \deg(\alpha)$.

If α is separable

$$r_1'(x) \neq 0$$

and hence has a finite number of roots S. We may add a constraint on the choice of Q saying that $a \notin r_1(S)$. Then since $r_1(x_0) = a$

$$x_0 \notin S$$

so $p - aq$ will not have repeated roots, i.e. $\#\ker(\alpha) = \deg(\alpha)$.

$$r_1'(x) = \frac{p(x)q'(x) - q'(x)p(x)}{q(x)^2}$$

Weil Pairing

Recall $\gcd(n, \mathrm{char}K) = 1$. For $Q \in E[n]$ take $f_Q \in K(E) : \mathrm{div}(f_Q) = n[Q] - n[\infty]$, there exists $g_Q \in K(E) : \mathrm{div}(g_Q^n) = \mathrm{div}(f_Q \circ [n])$.

For arbitrary $S \in E(K), P \in E[n]$

$$e_n(P, Q) = \frac{g_Q(S + P)}{g_Q(S)}$$

(this does not depend on the choice of S)

$$e_n : E[n] \times E[n] \to \mu_n(K)$$

$e_n(\alpha(P), \alpha(Q)) = e_n(P, Q)^{\deg \alpha}$

Let $\alpha : E \to E$ be a separable endomorphism.

Observe that $\alpha(P), \alpha(Q) \in E[n]$ since

$$n\alpha(P) = \alpha(nP) = \alpha(\infty) = \infty$$

Let $\{T_1, ..., T_k\} = \ker(\alpha)$. Since α is separable, $k = \deg(\alpha)$.

$$\mathrm{div}(f_Q) = n[Q] - n[\infty]$$
$$\mathrm{div}(f_{\alpha(Q)}) = n[\alpha(Q)] - n[\infty]$$
$$g_Q^n = f_Q \circ [n]$$
$$g_{\alpha(Q)}^n = f_{\alpha(Q)} \circ [n]$$

Let $\tau_T : E \to E$ be $X \to X + T$ translation by T.

Then $\mathrm{div}(f_Q \circ \tau_{-T_i}) = n[Q + T_i] - n[T_i]$.

Now notice that $\mathrm{div}(f_{\alpha(Q)}) = n[\alpha(Q)] - n[\infty]$ and so

$$\begin{aligned}\mathrm{div}(f_{\alpha(Q)} \circ \alpha) &= n \sum_{Q'' : \alpha(Q'') = \alpha(Q)} [Q''] - n \sum_{T : \alpha(T) = \infty} [T] \\ &= n \sum_{i=1}^{k} ([Q + T_i] + [T_i]) \\ &= \mathrm{div}(\prod_{i=1}^{k} f_Q \circ \tau_{-T_i})\end{aligned}$$

For $1 \leq i \leq k$ choose $T_i' \in E[n^2] : nT_i' = T_i$ then

$$\begin{aligned} g_Q(S - T_i')^n &= f_Q \circ [n](S - T_i') \\ &= f_Q(nS - T_i) \end{aligned}$$

by the definition of g_Q.

Now using this identity, we can see that

$$\begin{aligned} \operatorname{div}(\prod_{i=1}^{k} (g_Q \circ \tau_{-T_i'})^n) &= \operatorname{div}(\prod_{i=1}^{k} f_Q \circ \tau_{-T_i} \circ [n]) \\ &= \operatorname{div}(f_{\alpha(Q)} \circ \alpha \circ [n]) \end{aligned}$$

where we use the expression from above for $\operatorname{div}(f_{\alpha(Q)} \circ \alpha)$.

Notice $\alpha \circ [n] = [n] \circ \alpha$ because $n\alpha(P) = \alpha(nP)$, so multiplication by n commutes with endormorphisms.

$$\begin{aligned} \operatorname{div}(f_{\alpha(Q)} \circ \alpha \circ [n]) &= \operatorname{div}(f_{\alpha(Q)} \circ [n] \circ \alpha) \\ &= \operatorname{div}((g_{\alpha(Q)}^n) \circ \alpha) \\ &= \operatorname{div}((g_{\alpha(Q)} \circ \alpha)^n) \end{aligned}$$

Finally we get

$$\prod_{i=1}^{k} (g_Q \circ \tau_{-T_i'}) = g_{\alpha(Q)} \circ \alpha$$

$$\begin{aligned} e_n(\alpha(P), \alpha(Q)) &= \frac{g_{\alpha(Q)}(\alpha(P) + \alpha(S))}{g_{\alpha(Q)}(\alpha(S))} \\ &= \prod_{i=1}^{k} \frac{g_Q(P + S - T_i')}{g_Q(S - T_i')} \\ &= \prod_{i=1}^{k} e_n(P, Q) = e_n(P, Q)^k \\ &= e_n(P, Q)^{\deg \alpha} \end{aligned}$$

General Direction

$$\begin{aligned} \#E(\mathbb{F}_q) &= \#\ker(\Phi_q - \mathrm{id}) \\ &= \deg(\Phi_q - \mathrm{id}) \end{aligned}$$

then we can estimate this degree.

Separable Map

Definition of separable map

$$\deg \alpha = \#\ker(\alpha)$$

alternatively $r_1'(x) \neq 0$.

$P, Q \in E[n]$ and α is separable then $e_n(\alpha(P), \alpha(Q)) = e_n(P, Q)^{\deg \alpha}$.

Invariance of Weil Pairing under "action of Galois group"

$$\mathrm{Gal}(\bar{K}/K) = \{\sigma \in \mathrm{Aut}(\bar{K}) : \sigma|_k = \mathrm{id}_K\}$$

$$\Phi_q \in \mathrm{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$$

Proposition

$$\sigma \in \mathrm{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$$

$$\sigma(e_n(P,Q)) = e_n(\sigma P, \sigma Q)$$

Note $\sigma P \in E$ since $\sigma(y)^2 = \sigma(x)^3 + A\sigma(x) + B$, and then adding is rational so $P \in E[n] \Rightarrow n \cdot \sigma P = \infty$.
Recall that $f_Q, g_Q \in K(E)$

$$\mathrm{div}(f_Q) = n[Q] - n[\infty]$$

and g_Q that satisfy

$$g_Q^n = f_Q \circ [n]$$

and for any $S \in E(K)$

$$e_n(P,Q) = \frac{g_Q(P+S)}{g_Q(S)}$$

Write out f_Q and then when it equals zero, applying σ you see that σQ is now a root of f_Q^σ, so

$$\mathrm{div}(f_Q^\sigma) = n[\sigma Q] - n[\infty]$$

and similarly for g_Q^σ.

$$\begin{aligned}(g_Q^\sigma)^n &= (g_Q^n)^\sigma \\ &= (f_Q \circ [n])^\sigma \\ &= f_Q^\sigma \circ [n]\end{aligned}$$

Thus

$$\begin{aligned}\sigma(e_n(P,Q)) &= \sigma(\frac{g_Q(P+S)}{g_Q(S)}) \\ &= \frac{g_Q^\sigma(\sigma P + \sigma S)}{g_Q^\sigma(\sigma S)} \\ &= e_n(\sigma P, \sigma Q)\end{aligned}$$

Where the last step comes from the construction of the Weil pairing. Namely $g_Q^\sigma = g_{\sigma Q}$.

$$\begin{aligned}(g_{\sigma Q})^n &= f_{\sigma Q} \circ [n] \\ &= f_Q^\sigma \circ [n] \\ &= (g_Q^n)^\sigma \\ &= (g_Q^\sigma)^n \\ &= (f_Q \circ [n])^\sigma \\ &= f_Q^\sigma \circ [n]\end{aligned}$$

Restriction of α to $E[n]$ stays in $E[n]$

$$E[n] = \mathbb{Z}_n \times \mathbb{Z}_n$$

so $E[n] = \langle T_1, T_2 \rangle$.

$$\alpha_n = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\begin{aligned}\alpha(T_1) &= aT_1 + cT_2 \\ \alpha(T_2) &= bT_1 + dT_2 \\ \alpha(P) &= \alpha(rT_1 + sT_2) \\ &= r\alpha(T_1) + s\alpha(T_2)\end{aligned}$$

$$P = rT_1 + sT_2$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} r \\ s \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$
$$\alpha(P) = xT_1 + yT_2$$

$\det(\alpha_n) = \deg(\alpha) \mathbf{mod} n$

By weil pairing property $e_n(T_1, T_2)$ maps to a generator for $\mu_n(\mathbb{F}_q)$. Let $\eta = e_n(T_1, T_2)$. Since α is separable of Φ_q

$$\eta^{\deg(\alpha)} = e_n(T_1, T_2)^{\deg(\alpha)} = e_n(\alpha(T_1), \alpha(T_2))$$

But using the matrix we get

$$\begin{aligned} e_n(aT_1 + cT_2, bT_1 + dT_2) &= e_n(T_1, T_1)^{ab} e_n(T_1, T_2)^{ad} e_n(T_2, T_1)^{bc} e_n(T_2, T_2)^{cd} \\ &= 1^{ab} e_n(T_1, T_2)^{ad} e_n(T_2, T_1)^{bc} 1^{cd} \\ &= 1^{ab} e_n(T_1, T_2)^{ad} e_n(T_1, T_2)^{-bc} 1^{cd} \qquad \text{by pairing rule about swapping args} \\ &= e_n(T_1, T_2)^{ad-bc} \\ &= e_n(T_1, T_2)^{\det(\alpha_n)} \\ &= \eta^{\det(\alpha_n)} \end{aligned}$$

since η is a generator, we must have

$$\deg(\alpha) \equiv \det(\alpha_n) \mod n$$

$\deg(a\alpha + b\beta) = a^2 \deg(\alpha) + b^2 \deg(\beta) + ab(\deg(\alpha + \beta) - \deg(\alpha) - \deg(\beta))$

Restrict α, β using matrices α_n, β_n, where char $K \nmid n$.

Note from linear algebra matrix determinant rules $\det(a\alpha_n + b\beta_n) = a^2 \det(\alpha_n) + b^2 \det(\beta_n) + ab(\det(\alpha_n + \beta_n) - \det(\alpha_n) - \det(\beta_n))$.

Now replace determinant by degree for mod n.

Since this is true for infinitely many n's, we have ordinary equality.

$\deg(r\Phi_q + s) = r^2 q + s^2 - rst$

$r, s \in \mathbb{Z}, \gcd(s, q) = 1$ then

$$t = q + 1 - \deg(\Phi_q - 1)$$

By previous proposition

$$\deg(r\Phi_q - s) = r^2 \deg(\Phi_q) + s^2 \deg(-1) + rs(\deg(\Phi_q - 1) - \deg(\Phi_q) - \deg(-1))$$

Since $\deg(\Phi_q) = q$ and $\deg(-1) = 1$

$$\deg(r\Phi_q - s) = r^2 q + s^2 + rs(\deg(\Phi_q - 1) - q - 1)$$

Hasse-Weil Theorem

$$|q + 1 - \#E(\mathbb{F}_q)| \leq 2\sqrt{q}$$

$$\deg(\Phi_q - 1) = \#\ker(\Phi_q - 1) = \#E(\mathbb{F}_q)$$

For any $r, s \in \mathbb{Z}$ such that $\gcd(s, q) = 1$, we have

$$0 \leq \deg(r\Phi_q - s)$$

because degrees are greater than 0.

$$r^2q + s^2 - rst >= 0$$
$$\Leftrightarrow q(\frac{r}{s})^2 - t(\frac{r}{s}) + 1 \geq 0$$

The set of all rational numbers r/s such that $\gcd(s, q) = 1$ is dense in $\mathbb{R}$ so the polynomial

$$qx^2 - tx + 1$$

gets only non-negative values, and has non-positive discriminant.

$$t^2 - 4q \leq 0$$

Dense Set

If $\forall x \in \mathbb{R}$, there exists a sequence

$$s_1, s_2, ..., s_n, ...$$
$$\lim_{n\to\infty} s_n = x$$

For example π can be approximated with an infinite sequence of rationals.

Take $x_0 \in \mathbb{R}$ since there exists a sequence $\sigma_n = r_n/s_n$ such that $\lim \sigma_n = x_0$.

$$0 \leq \lim_{n\to\infty}(q\sigma_n^2 - t\sigma_n + 1) = q(\lim_{n\to\infty})^2 - t \lim_{n\to\infty}(\sigma_n) + 1$$
$$\Rightarrow qx_0^2 - tx_0 + 1 \geq 0$$

Hasse-Weil Corrollary

In $\mathrm{End}(E)$

$$\Phi_q^2 - [t] \circ \Phi_q + [q] = 0$$

For all $p \nmid n$

$$E[n] \cong \mathbb{Z}_n \times \mathbb{Z}_n$$

so we represent $\Phi_q|_{E[n]} : E[n] \to E[n]$ as a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ (choose generators $\{T_1, T_2\} \subseteq E[n]$ which correspond to $\{(1, 0), (0, 1)\} \in \mathbb{Z}_n \times \mathbb{Z}_n$)

Any $2x2$ satisfies

$$A^2 - \mathrm{tr}(A)A + \det(A)I = 0$$

where $\mathrm{tr}(A) = a + d$.

We showed that

$$\det(A_n) = \deg(\Phi_n) \mod n$$

and another direct calc shows

$$\mathrm{tr}(A_n) = 1 + \det(A_n) - \det(I - A_n)$$

thus

$$\begin{aligned}\mathrm{tr}(A_n) &= 1 + \deg\Phi_q + \deg(\mathrm{id} - \Phi_q) \\ &= 1 + \deg\Phi_q + \deg(\Phi_q - \mathrm{id}) \\ &= 1 + q - (q + 1 - t) \mod n \\ &= t \mod n\end{aligned}$$

substititng, we get

$$A^2 - t \cdot A + q \cdot I = 0$$

Now since this is true for infinitely many n, it should be true in $\mathrm{End}(E) \Rightarrow$

$$\Phi_q^2 + [t] \cdot \Phi_q + [q] = 0$$